MIE312 Fluid Mechanics

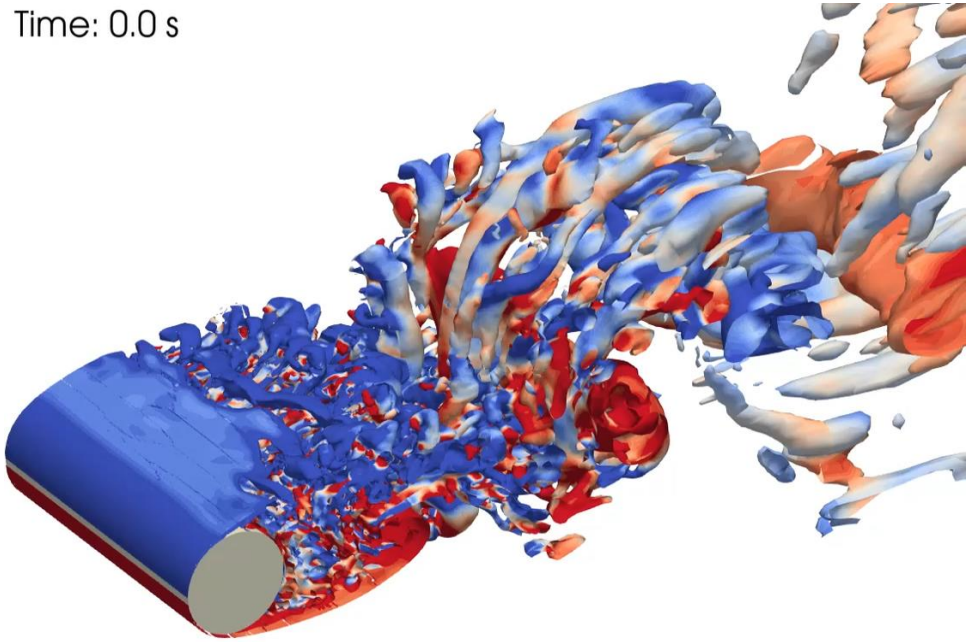
Tutorial 1 – Ch.1 – Viscosity

Howard Ho

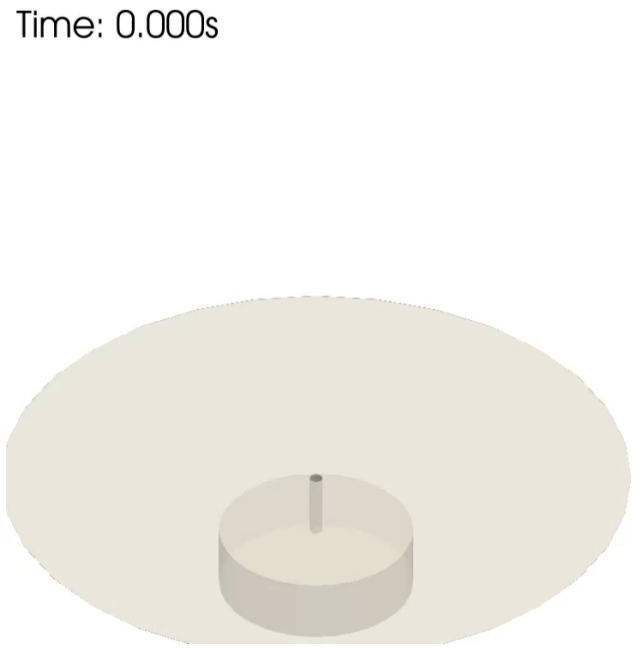


TA Introduction





Turbulent flow over a circular cylinder



Synthetic jets in quiescent

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Courses TAed: MIE 100 (2021-23, 2025-26); Course Instructor (2024)

MIE 312 (2024, 2026)

MIE 507 (2025-2026)

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Tutorial Schedule/Structure

- 9 Tutorials in total
- Every Thursday 6-7 PM, 2 parallel sessions
- TA Rojin runs the other tutorial
- 15-20min recap + 2-3 Example problems
- Preview slides uploaded by Monday 9am
- Annotated version uploaded by 6pm the day after each tutorial
- Anonymous feedback form around Tutorial 3 or 4
- Tentative office hours for mid terms

Supplementary Slides

Curved Surface - Pressure (Ch.2)

Bouyancy (Ch.2)

Tutorial Week	Topics (2026)
1	Viscosity (Ch.1)
2	Surface Tension (Ch.1)
3	Hydrostatic Pressure (Ch.2)
	Manometer (Ch.2)
4	Mid-Term 1
5	RTT - Mass (Ch.3)
	RTT - Linear Momentum (Ch.3)
6	Bernoulli Equation (Ch.3)
	RTT - Angular Momentum (Ch.3)
X	(Reading Week)
7	Differential Relations (Ch.4)
	Dimensional Analysis (Ch.5)
8	Mid-Term 2
9	Moving CV, non-steady (Ch.3)
	RTT - Energy (Ch.3)
10	Moody Chart - Pipe Flow (Ch.6)
	Head Loss (Ch.6)
11	External Flow (Ch.7)
	Drag Coefficient (Ch.7)

*Schedule is tentative, minor adjustment might occur over the semester

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Crib Sheet

Midterms and Final Exam

All material covered in the course is examinable, including lectures, tutorials, Connect homework, laboratories, and concepts reinforced through optional practice problems. Optional practice problems are not graded, but they reflect concepts and problem types that may be examined.

- Midterm 1: Thursday, October 8, 2026. Time, format, and room distribution will be posted on Quercus.
- Midterm 2: Thursday, November 12, 2026. Time, format, and room distribution will be posted on Quercus.
- Final exam: scheduled by the Faculty during the December 10-22, 2026 Engineering final exam period.
- Non-programmable calculators are allowed for the midterm and final exams. Calculators must be non-printing, non-communicating, silent, and self-powered.
- Students will be provided with a course crib/formula sheet for midterms and the final exam. Students may not bring their own crib sheets or formula sheets.

From Quercus/Syllabus

You won't need page 3 until Ch.6 (Tutorial – 10)

MIE 312 – Fluid Mechanics I: Midterm 2 and Final Crib Sheet

1. PROPERTIES & CONVERSIONS

Ideal Gas: $P = \rho RT$; $R_{air} = 287 \frac{J}{kgK} = 1716 \frac{ft\cdot lb_f}{slug\cdot R}$; $\gamma = \rho g$; $g = 9.81\ m/s^2$

Viscosity: $\tau = \mu \frac{du}{dy}$; $\nu = \mu/\rho$; 1 Poise = 0.1 Pa·s; $F_\tau = \tau A$

Bulk Modulus: $E_v = \frac{-dp}{dV/V}$; Sound: $a = \sqrt{kRT}$

Surface Tension: $\Delta P_{drop} = 2\Upsilon/R$; $\Delta P_{bub} = 4\Upsilon/R$; $h = \frac{2\Upsilon\cos\theta}{\rho g R}$

Capillarity Error: $\%\text{error} = \frac{\text{Measured} - \text{True}}{\text{True}} \times 100\%$

Moment: $\mathbf{M} = \mathbf{r} \times \mathbf{F}$; $|M| = Fd_\perp$

Imperial: $g = 32.2 \frac{ft}{s^2}$; $\rho_w = 1.94 \frac{slug}{ft^3}$; $\gamma_w = 62.4 \frac{lb_f}{ft^3}$

1 slug = 14.59 kg; 1 hp = 550 $\frac{ft\cdot lb_f}{s}$ = 746 W

1 atm = 14.7 psi = 2116 psf = 101.3 kPa

Rheology: $\tau = K(\dot{\gamma})^n$ ($n < 1$ thin; $n > 1$ thick)

2. HYDROSTATICS & ATMOSPHERE

Statics: $P_2 - P_1 = -\gamma(z_2 - z_1)$; HGL: $HGL = z + P/\gamma$

Isothermal: $P = P_1 \exp[-g\Delta z/RT]$

Troposphere: $P = P_0(1 - \beta z/T_0)^{g/R\beta}$ ($\beta = 0.0065$ K/m)

Hydrostatic: $P = P_0 + \rho g h$

Pascal Law: $F = P \times A$; $P_1 = P_2$

Planes: $F = \gamma h_{CG} A$; $y_{cp} = y_{CG} + \frac{I_{xx,CG}\sin\theta}{h_{CG} A}$

Curved: $F_H = \gamma h_{CG} A_{proj}$; F_V = Weight of fluid above

Buoyancy: $F_B = \rho g \forall_{sub}$

Stability: $\overline{MG} = \frac{I_{xx}}{\forall_{sub}} - \overline{GB}$ (Stable if $\overline{MG} > 0$)

Rigid Body: $\nabla P = \rho(\vec{g} - \vec{a})$; Rotate: $z = \frac{\omega^2 r^2}{2g} + C$

Moving manometer: $A_1 a = A_2 b$; inclined-leg vertical change = $b \sin\theta$; write initial/final pressure walks and subtract.

3. CONSERVATION LAWS (CV)

RTT: $\frac{dB_{sys}}{dt} = \frac{\partial}{\partial t}\int_{CV} \rho\beta d\forall + \int_{CS} \rho\beta(\mathbf{V_r}\cdot\mathbf{n})dA$

Relative Vel: $\mathbf{V_r} = \mathbf{V_f} - \mathbf{V_{cv}}$; $\dot{m}_{rel} = \rho A V_r$

Rotating outlet: $\mathbf{V} = \mathbf{W} + \mathbf{U}$; $\mathbf{U} = \boldsymbol{\omega} \times \mathbf{r}$

Steady 1-D Ang. Mom.: $M_z = \sum_{out} \dot{m} r V_\theta - \sum_{in} \dot{m} r V_\theta$

Mass: $\sum \dot{m}_{in} = \sum \dot{m}_{out} + \frac{\partial m_{cv}}{\partial t}$

Momentum: $\sum \mathbf{F} = \frac{\partial}{\partial t}\int \mathbf{V}\rho d\forall + \sum(\dot{m}\mathbf{V})_{out} - \sum(\dot{m}\mathbf{V})_{in}$

Ang. Mom (Stationary): $\sum \mathbf{M}_{ext} = \frac{\partial}{\partial t}\int_{CV}(\mathbf{r}\times\mathbf{V})\rho d\forall + \int_{CS}(\mathbf{r}\times\mathbf{V})\rho(\mathbf{V}\cdot\mathbf{n})dA$

Momentum Flux (Vanes): $F_x = \dot{m}(V_{1,x} - V_{2,x})$; $F_y = \dot{m}(V_{1,y} - V_{2,y})$

Energy: $\frac{p_1}{\gamma} + \alpha_1\frac{V_1^2}{2g} + z_1 + h_p = \frac{p_2}{\gamma} + \alpha_2\frac{V_2^2}{2g} + z_2 + h_t + h_L$

Factors: $\alpha = 2.0$ (Lam), $\alpha \approx 1.05$ (Turb)

Grade Lines: $EGL = HGL + \alpha V^2/2g$

4. PITOT TUBE & PRESSURE TYPES

Static Pressure (P): Actual thermodynamic pressure.

Stagnation (P_0): Pressure when $V \to 0$ isentropically.

Dynamic Pressure: $q = \frac{1}{2}\rho V^2$

Relation: $P_0 = P + \frac{1}{2}\rho V^2 \to V = \sqrt{\frac{2(P_0 - P)}{\rho}}$

Pitot-Static Tube: $\Delta P = h(\gamma_m - \gamma_f) \to V = \sqrt{2gh(\rho_m/\rho_f - 1)}$

5. TURBULENT FRICTION RELATIONS (APPROX)

Colebrook–White (Implicit): $\frac{1}{\sqrt{f}} = -2.0\log_{10}\left[\frac{\epsilon/D}{3.7} + \frac{2.51}{Re\sqrt{f}}\right]$

Haaland (Approx): $\frac{1}{\sqrt{f}} \approx -1.8\log_{10}\left[\left(\frac{\epsilon/D}{3.7}\right)^{1.11} + \frac{6.9}{Re}\right]$

Swamee-Jain (f only): $f \approx 0.25/\left[\log_{10}\left(\frac{\epsilon/D}{3.7} + \frac{5.74}{Re^{0.9}}\right)\right]^2$

6. INTERNAL PIPE FLOW BASICS

Head Loss: $h_L = \left(f\frac{L}{D} + \sum K_L\right)\frac{V^2}{2g}$

Power: $P = \omega T_0$; $P = \gamma Q h$

Pump Efficiency: $\eta = \frac{P_{rwhp}}{P_{input}}$

Laminar ($Re < 2300$): $f = 64/Re$; $Q = \frac{\pi D^4 \Delta P}{128\mu L}$

Entry: $L_{e,lam} \approx 0.06 ReD$; $L_{e,turb} \approx 4.4 Re^{1/6} D$

Non-Circular: $D_h = 4A/P_w$; Junctions: $\sum Q_{node} = 0$

Hardy Cross: $\Delta Q = -\frac{\sum h_L}{2\sum(K_L/Q)}$

7. MINOR LOSSES (K_L)

Ent. (Sharp)	0.5	Elbow (90°)	0.9
Ent. (Rnd)	0.04	Gate (Open)	0.2
Exit (All)	1.0	Globe (Open)	10
Check Valve	2.0	Tee (Branch)	1.8

8. BOUNDARY LAYERS & EXTERNAL

Params: $\delta^* = \int(1 - \frac{u}{U})dy$; $\theta = \int \frac{u}{U}(1 - \frac{u}{U})dy$

Karman: $\frac{C_f}{2} = \frac{d\theta}{dx} + (H+2)\frac{\theta}{U}\frac{dU}{dx}$ ($H = \delta^*/\theta$)

Flat Plate:

Flow	δ/x	$C_{f,x}$	C_D
Lam.	$5/\sqrt{Re_x}$	$0.664/\sqrt{Re_x}$	$1.328/\sqrt{Re_L}$
Turb.	$0.16/Re_x^{1/7}$	$0.027/Re_x^{1/7}$	$0.031/Re_L^{1/7}$

Laminar displacement: $\delta^* = 1.7208x/\sqrt{Re_x} \approx 0.344\delta$

Separation: $\lambda = \frac{\theta^2}{\nu}\frac{dU}{dx}$; $\lambda_{sep} \approx -0.09$; $H_{sep} \approx 3.5$

Howarth: $x_{sep}/L \approx 0.12$; Görtler: $x_{sep}/L \approx 0.16$

Drag: $F_D = C_D A \frac{1}{2}\rho V^2$; Sphere: $C_D = 24/Re$ (Stokes)

Falling sphere: $U_t = \sqrt{\frac{4}{3}\frac{g_0 - \rho}{\rho}\frac{gD}{C_D}}$; iterate with $Re = U_t D/\nu$.

MIE 312 – Fluid Mechanics I: Midterm 2 and Final Crib Sheet (continued)

9. DRAG COEFFICIENTS (C_D)

Square Rod	2.0	Circular Disk	1.17
Cyl (long)	1.2	Hemisph (hol)	1.33
Streamlined	0.04	Sphere (turb)	0.45

10. EGL/HGL DRAWING RULES

1. **EGL always slopes down** in direction of flow.
2. **HGL is always** $aV^2/2g$ **below EGL.**
3. **Pump:** Sudden jump in both; **Turbine:** Sudden drop.
4. **Nozzle:** Gap between EGL and HGL increases ($V \uparrow$).
5. **Free Jet:** $P_{gauge} = 0$, so HGL follows centerline.
6. **Reservoir:** EGL/HGL at free surface level.

11. FLOW MEASUREMENT

$Q = C_d A_1\sqrt{\frac{2\Delta P/\rho}{1-\beta^4}}$ ($\beta = d/D$)

Venturi: $C_d \approx 0.98$; Orifice: $C_d \approx 0.61$

12. DIMENSIONAL ANALYSIS

Pi Theorem: number of Π groups $= n - r$, where n is the number of variables and r is the dimensional-matrix rank. Repeating variables must be dimensionally independent.

$Re = \frac{\rho V L}{\mu}$; $Fr = \frac{V}{\sqrt{gL}}$; $Ma = \frac{V}{a}$; $We = \frac{\rho V^2 L}{\Upsilon}$

$Eu = \frac{\Delta P}{\rho V^2}$; $St = \frac{\omega L}{V}$; $Ca = \frac{\mu V}{\Upsilon}$

$\Pi_\tau = \frac{\tau}{\rho V^2}$; $f_D = 8\Pi_\tau$; laminar $\Pi_\tau = 8/Re$;

Blasius $\Pi_\tau \approx 0.0396 Re^{-1/4}$; $\sigma_{cav} = \frac{P - P_v}{\frac{1}{2}\rho V^2}$; $C_p = \frac{P - P_\infty}{\frac{1}{2}\rho V_\infty^2}$

Similarity: $\Pi_{model} = \Pi_{prototype}$

6B. DIFFERENTIAL FLUID MOTION

2-D field: $\mathbf{V} = u(x,y)\mathbf{i} + v(x,y)\mathbf{j}$

Normal strain: $\dot{\epsilon}_{xx} = \frac{\partial u}{\partial x}$; $\dot{\epsilon}_{yy} = \frac{\partial v}{\partial y}$

Angular deformation: $\dot{\gamma}_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}$

Vorticity: $\zeta = \nabla \times \mathbf{V}$; $\zeta_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$

Element rotation: $\Omega = \frac{1}{2}\zeta$

2-D incompressible: $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$

No slip at a wall: $\mathbf{V}_{fluid} = \mathbf{V}_{wall}$; use the stated profile form and wall conditions to determine constants

13. TURBOMACHINERY

Euler: $H = \frac{U_2 V_{t2} - U_1 V_{t1}}{g}$; $T = \dot{m}(r_2 V_{t2} - r_1 V_{t1})$

System: $H_{req} = \Delta z + \Delta P/\gamma + h_L(Q)$

Affinity: $\frac{Q_1}{Q_2} = \frac{N_1}{N_2}\left(\frac{D_1}{D_2}\right)^3$; $\frac{H_1}{H_2} = \left(\frac{N_1}{N_2}\right)^2\left(\frac{D_1}{D_2}\right)^2$

Dimensionless Spec. Speed: $N_s^* = \frac{\omega\sqrt{Q}}{(gH)^{3/4}}$

NPSH: $H_{atm} + H_{stat} - H_{vap} - h_{loss}$

14. GEOMETRIC PROPERTIES

Shape	Area	Centroid	Inertia (I_{xc})
Rectangle	bh	$h/2$	$bh^3/12$
Circle	πR^2	R	$\pi R^4/4$
Semi-Circ	$\pi R^2/2$	$4R/3\pi$	$0.1098R^4$
Quart-Circ	$\pi R^2/4$	$4R/3\pi$	$0.0549R^4$
Triangle	$bh/2$	$h/3$	$bh^3/36$

15. NOMENCLATURE & DIMS

m	Mass	kg	M
A, D	Area, Dia	m^2, m	L^2, L
I	Second Moment of Area	m^4	L^4
v, a	Velo, Accel	$m/s, m/s^2$	LT^{-1}, LT^{-2}
F, T	Force, Torque	N, Nm	$ML/T^2, ML^2/T^2$
$P, \dot{W}$	Pressure, Power	Pa, W	$M/LT^2, ML^2/T^3$
γ, ρ	Sp Weight, Dens	$N/m^3, kg/m^3$	$M/L^2T^2, M/L^3$
μ, ν	Viscosity	$Pa\cdot s, m^2/s$	$M/LT, L^2/T$
τ	Shear Stress	Pa	$ML^{-1}T^{-2}$
Υ	Surf. Tension	N/m	MT^{-2}

16. WATER PROPERTIES VS. TEMP

T (°C)	ρ (kg/m³)	γ (kN/m³)	μ (mPa·s)	P_{vap} (kPa)
0	1000	9.81	1.79	0.61
20	998	9.79	1.00	2.34
40	992	9.73	0.65	7.38
60	983	9.65	0.47	19.9
80	972	9.53	0.35	47.3
100	958	9.40	0.28	101.3

17. MATH & MISC

Quadratic: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ **Log:** $\ln(a/b) = \ln a - \ln b$

Vectors: $\vec{A}\cdot\vec{B} = AB\cos\theta$; $|\vec{A}\times\vec{B}| = AB\sin\theta$

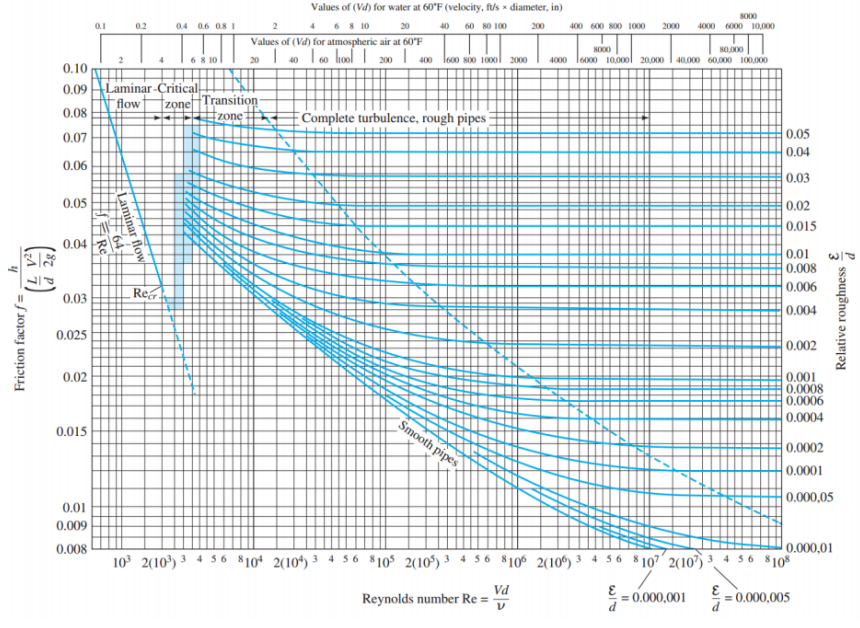
Stagnation: $\frac{P_0}{P} = (1 + \frac{k-1}{2}Ma^2)^{k/(k-1)}$

18. ITERATIVE ALGORITHMS

Type 2 (Find Q): Guess $f \to V = \sqrt{\frac{2gh_L}{fL/D + \sum K_L}} \to Re \to f_{new}$ from Haaland.

Type 3 (Find D): Guess f; use $V = 4Q/(\pi D^2)$ and solve $h_{avail} = (fL/D + \sum K_L)V^2/(2g)$ for D; then $Re = \rho V D/\mu$, $\epsilon/D \to f_{new}$.

Moody Friction Factor Chart



Common Greek Symbols

$A\ \alpha$	$B\ \beta$	$\Gamma\ \gamma$	$\Delta\ \delta$	$E\ \epsilon$	$Z\ \zeta$
Alpha	**Beta**	**Gamma**	**Delta**	**Epsilon**	**Zeta**
$H\ \eta$	$\Theta\ \theta$	$I\ \iota$	$K\ \kappa$	$\Lambda\ \lambda$	$M\ \mu$
Eta	**Theta**	**Iota**	**Kappa**	**Lambda**	**Mu**
$N\ \nu$	$\Xi\ \xi$	$O\ o$	$\Pi\ \pi$	$P\ \rho$	$\Sigma\ \sigma, \varsigma$
Nu	**Xi**	**Omicron**	**Pi**	**Rho**	**Sigma**
$T\ \tau$	$\Upsilon\ \upsilon$	$\Phi\ \phi$	$X\ \chi$	$\Psi\ \psi$	$\Omega\ \omega$
Tau	**Upsilon**	**Phi**	**Chi**	**Psi**	**Omega**

Latex math:

$$\tau = \mu \frac{du}{dy}$$

$$\Sigma \int \vec{V} \partial t$$

$$\omega \Omega$$

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Question Time



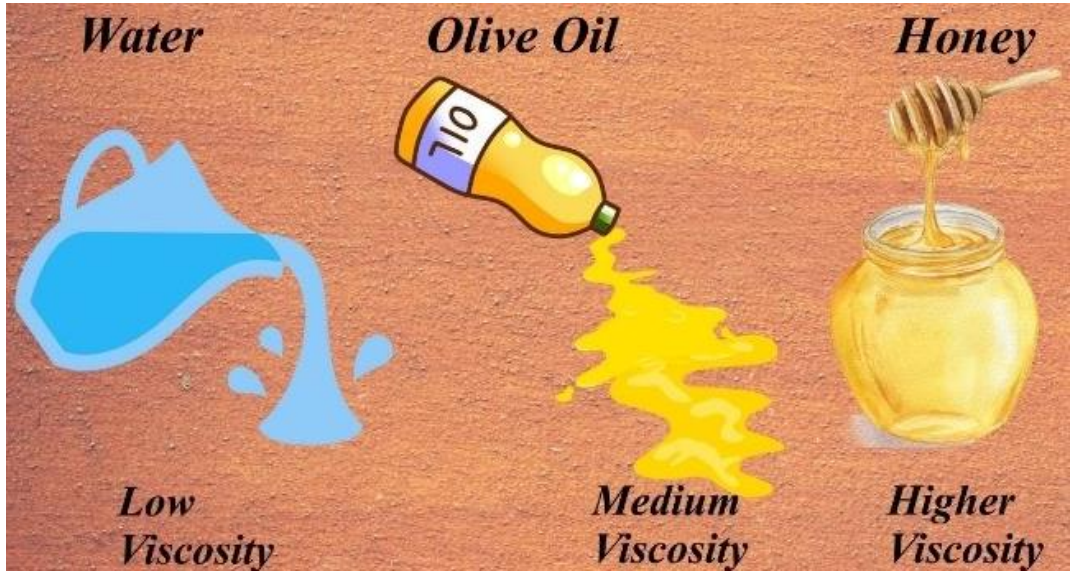
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Learning Objective

- Establish the basics on units and dimensions
- Understand the effect of viscosity
- Calculate shear stress from velocity gradient and fluid viscosity







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Units and Dimensions

Primary dimension	SI unit	BG unit	Conversion factor
Mass $\{M\}$	Kilogram (kg)	Slug	1 slug = 14.5939 kg
Length $\{L\}$	Meter (m)	Foot (ft)	1 ft = 0.3048 m
Time $\{T\}$	Second (s)	Second (s)	1 s = 1 s
Temperature $\{\Theta\}$	Kelvin (K)	Rankine (°R)	1 K = 1.8°R

Example (Newton's 2nd law):

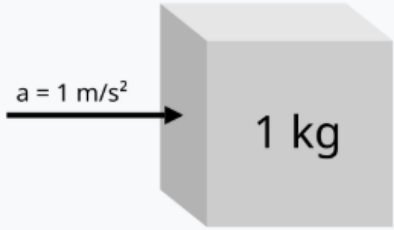
$$F = m\,a$$

$$F = kg\,.\,m/s^2$$

$$\therefore 1\,N = 1\,kg\,.\,m/s^2$$

newton

1 N



Visualization of one newton of force

General information

Unit system	SI
Unit of	force
Symbol	N
Named after	Sir Isaac Newton

Conversions

1 N *in ...*	*... is equal to ...*
SI base units	1 kg·m·s^{-2}
CGS units	10^5 dyn
Imperial units	0.224809 lbf

SI = International System
BG = British Gravitational

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Dimensional Analysis

Secondary dimension	SI unit	BG unit	Conversion factor
Area $\{L^2\}$	m^2	ft^2	$1\ m^2 = 10.764\ ft^2$
Volume $\{L^3\}$	m^3	ft^3	$1\ m^3 = 35.315\ ft^3$
Velocity $\{LT^{-1}\}$	m/s	ft/s	1 ft/s = 0.3048 m/s
Acceleration $\{LT^{-2}\}$	m/s^2	ft/s^2	$1\ ft/s^2 = 0.3048\ m/s^2$
Pressure or stress $\{ML^{-1}T^{-2}\}$	$Pa = N/m^2$	lbf/ft^2	$1\ lbf/ft^2 = 47.88\ Pa$
Angular velocity $\{T^{-1}\}$	s^{-1}	s^{-1}	$1\ s^{-1} = 1\ s^{-1}$
Energy, heat, work $\{ML^2T^{-2}\}$	$J = N \cdot m$	$ft \cdot lbf$	$1\ ft \cdot lbf = 1.3558\ J$
Power $\{ML^2T^{-3}\}$	$W = J/s$	$ft \cdot lbf/s$	$1\ ft \cdot lbf/s = 1.3558\ W$
Density $\{ML^{-3}\}$	kg/m^3	$slugs/ft^3$	$1\ slug/ft^3 = 515.4\ kg/m^3$
Viscosity $\{ML^{-1}T^{-1}\}$	$kg/(m \cdot s)$	$slugs/(ft \cdot s)$	$1\ slug/(ft \cdot s) = 47.88\ kg/(m \cdot s)$
Specific heat $\{L^2T^{-2}\Theta^{-1}\}$	$m^2/(s^2 \cdot K)$	$ft^2/(s^2 \cdot °R)$	$1\ m^2/(s^2 \cdot K) = 5.980\ ft^2/(s^2 \cdot °R)$

Primary dimension
Mass $\{M\}$
Length $\{L\}$
Time $\{T\}$
Temperature $\{\Theta\}$

$$F = m\,a$$

Unit: $kg.m/s^2$

Dimension:

$$F = M.L\,T^{-2}$$

Chapter 5

Dimensional Analysis and Similarity 285

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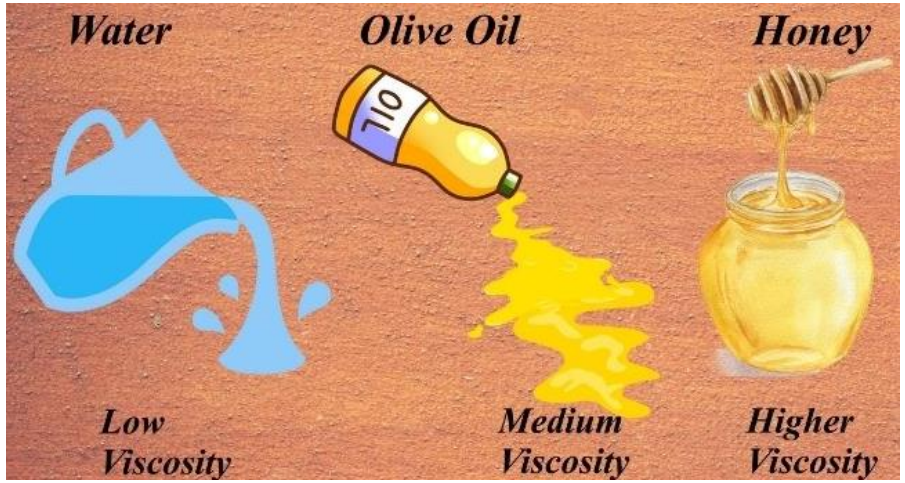
Viscosity

- Viscosity is a quantitative measure of a fluid's resistance to flow
- Air is less viscous than water
  - It's tougher to wave your arm under water than in air
- Viscosity is temperature dependent
  - Increasing temperature tends to decrease the viscosity

More viscous = Stronger resistance



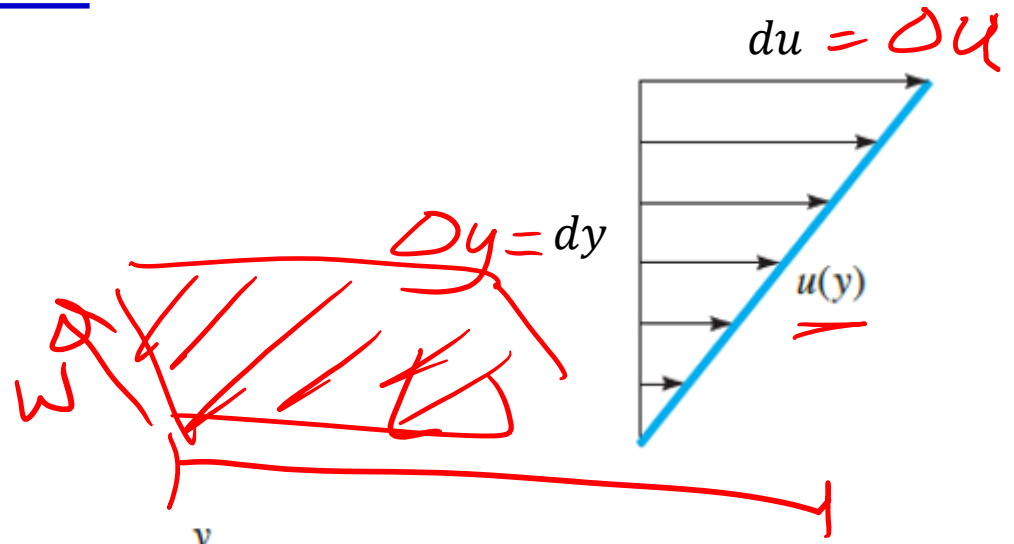


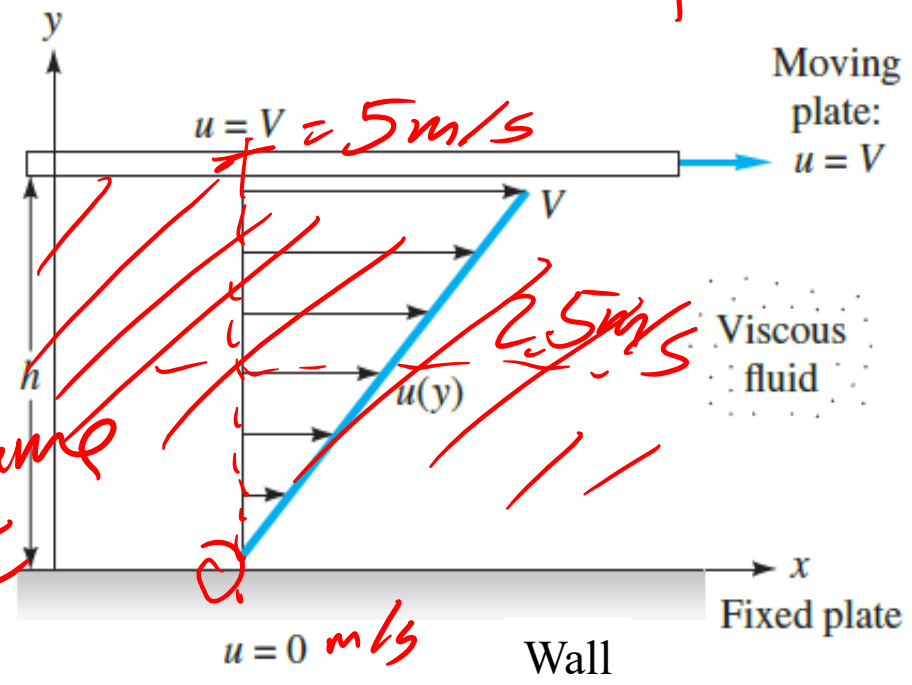


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Shear Stress τ

- Shear stress τ (Pascal, pronounced as "**tau**")
- $\tau = \mu \frac{du}{dy}$
- There are two types of viscosity
  - Dynamic viscosity μ (pronounced as "**mu**")
  - Kinematic viscosity ν (pronounced as "**nu**")
  - $\nu = \frac{\mu}{\rho}$
- If the velocity profile is linear and μ is constant, the shear stress τ is constant everywhere
- $F_{shear} = \tau A$





Shear stress	
Common symbols	τ
SI unit	pascal
Derivations from other quantities	$\tau = \frac{F}{A}$

ρ = density of fluid (pronounced as "rho")

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Question Time



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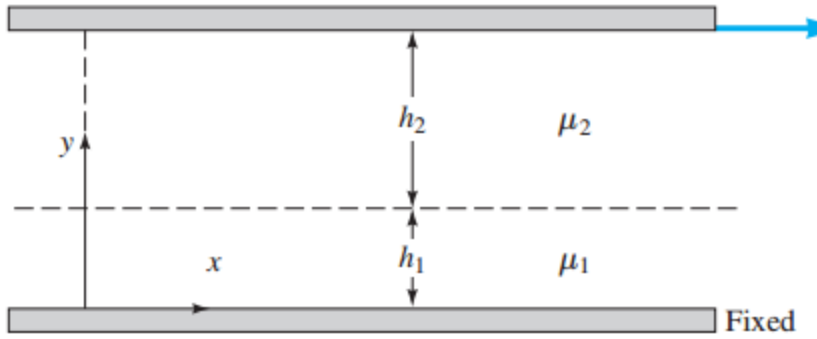
Problem 1

Examining the steady flow between a fixed lower plate and a moving upper plate, there are two immiscible liquids between the plates, as in Fig. P1.57.

(a) Sketch the expected no-slip velocity distribution u(y) between the plates.

(b) Find an analytic expression for the velocity U at the interface between the two liquid layers.

(c) What is the result of (b) if the viscosities and layer thicknesses are equal?



P1.57

1. PROPERTIES & CONVERSIONS

Ideal Gas: $P = \rho RT$; $R_{air} = 287\frac{J}{kgK} = 1716\frac{ft \cdot lb}{slug \cdot R}$;
$\gamma = \rho g$; $g = 9.81\ \text{m/s}^2$

Viscosity: $\tau = \mu\frac{du}{dy}$; $\nu = \mu/\rho$; 1 Poise = 0.1 Pa · s;
$F_\tau = \tau A$

Bulk Modulus: $E_v = \frac{-dP}{d\forall/\forall}$; **Sound:** $a = \sqrt{kRT}$

Surface Tension: $\Delta P_{drop} = 2\Upsilon/R$; $\Delta P_{bub} = 4\Upsilon/R$;
$h = \frac{2\Upsilon\cos\theta}{\rho g R}$

Capillarity Error: $\%\,\text{error} = \dfrac{\text{Measured} - \text{True}}{\text{True}} \times 100\%$

Moment: $\mathbf{M} = \mathbf{r} \times \mathbf{F}$; $|M| = Fd_\perp$

Imperial: $g = 32.2\frac{ft}{s^2}$; $\rho_w = 1.94\frac{slug}{ft^3}$; $\gamma_w = 62.4\frac{lb}{ft^3}$

1 slug = 14.59 kg; 1 hp = $550\frac{ft \cdot lb}{s}$ = 746 W

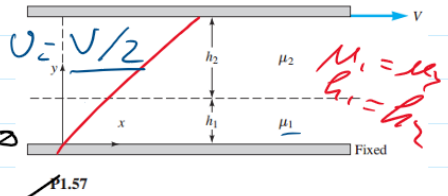
1 atm = 14.7 psi = 2116 psf = 101.3 kPa

Rheology: $\tau = K(\dot{\gamma})^n$ ($n < 1$ thin; $n > 1$ thick)

An immiscible liquid is incapable of mixing (e.g. oil always float above water)

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Examining the steady flow between a fixed lower plate and a moving upper plate, there are two immiscible liquids between the plates, as in Fig. P1.57.
(a) Sketch the expected no-slip velocity distribution u(y) between the plates.
(b) Find an analytic expression for the velocity U at the interface between the two liquid layers.
(c) What is the result of (b) if the viscosities and layer thicknesses are equal?



P1.57

Given, $\mu_1 < \mu_2$ $\quad \mu_2 < \mu_1$

(a) Shear stress τ is same everywhere. $\quad \tau_1 = \tau_2$

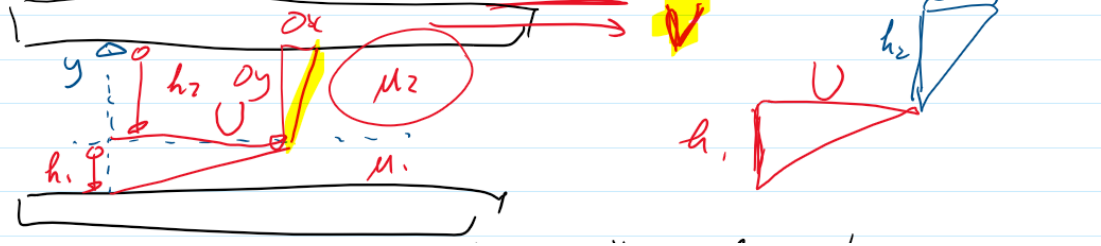
(cannot mix)

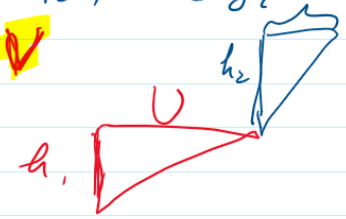
$\tau_1 = \tau_2$

$$\mu_1 \frac{du_1}{dy_1} = \mu_2 \frac{du_2}{dy_2}$$

If $\mu_2 = 10\,\mu_1$

$$\mu_1 \frac{du_1}{dy_1} = 10\,\mu_1 \frac{du_2}{dy_2}(V-U)$$





(b) Assume no-slip at walls, set velocity at interface to "U"

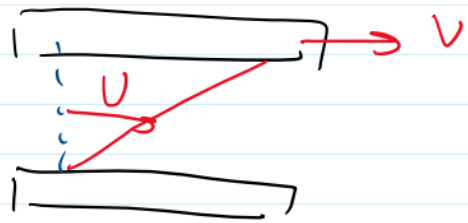
$\tau_1 = \tau_2$

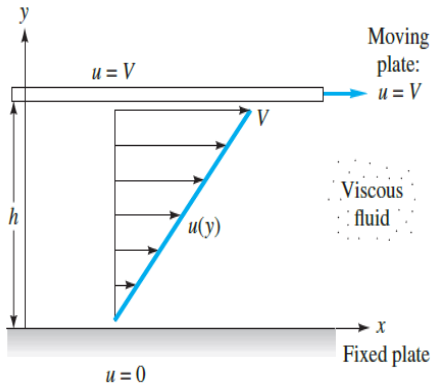
$$\mu_1 \left(\frac{U}{h_1}\right) = \mu_2 \left(\frac{V-U}{h_2}\right)$$

$$U\frac{\mu_1}{h_1} = V\frac{\mu_2}{h_2} - U\frac{\mu_2}{h_2}$$

$$U = V\left[\frac{\mu_2/h_2}{\mu_1 h_1 + \mu_2 h_2}\right]$$ ✓

(c) $U = V/2$





For part (c), we are told viscosity μ and layer thickness h are both same, the previous τ assumption still holds true, so the problem basically becomes like the recap slide's.

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Problem 2

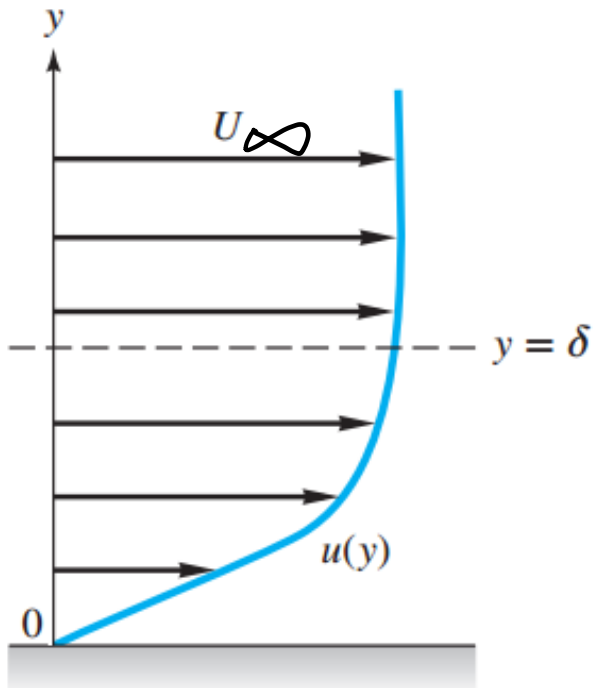
An approximation for the boundary-layer shape shown in the figure is the formula

$$u(y) \approx U \sin\left(\frac{\pi y}{2\delta}\right), 0 \leq y \leq \delta$$

where U is the stream velocity far from the wall and δ is the boundary layer thickness. If the fluid is helium at $20°\text{C}$ and $1\ \text{atm}$, and if $U = 10.8\ \text{m/s}$ and $\delta = 3\ \text{mm}$, use the formula to:

(a) Estimate the wall shear stress τ_w in Pa.

(b) Find the position in the boundary layer where τ is one-half of τ_w.



$$\mu = 1.97 \times 10^{-5}$$

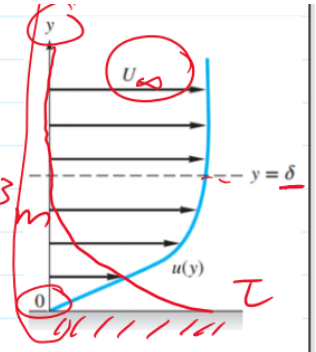
$$\tau = \mu \frac{du}{dy}$$

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An approximation for the boundary-layer shape shown in the figure is the formula $u(y) \approx U \sin\left(\frac{\pi y}{2\delta}\right)$, $0 \le y \le \delta$, where U is the stream velocity far from the wall and δ is the boundary layer thickness. If the fluid is helium at 20°C "an 1 atm", and if $U = 10.8$ m/s"and $\delta = 3$ mm", use the formula to:

(a) Estimate the wall shear stress τ_w in Pa.

(b) Find the position in the boundary layer where τ is one-half of τ_w.



$\delta = 3 \times 10^{-3}$ m

(a) τ_{wall} → wall shear stress @ $y = 0$

$\tau = \mu \frac{du}{dy}$

$\frac{d}{dy}(u(y)) = U\left(\frac{\pi}{2\delta}\right)\cos\left(\frac{\pi y}{2\delta}\right)$

$\mu = 1.97 \times 10^{-5}$

$\tau = \mu \frac{du}{dy}$

$\tau = \mu U\left(\frac{\pi}{2\delta}\right)\cos\left(\frac{\pi y}{2\delta}\right)$ — (1)

sub $y = 0$ m, $U = 10.8$ m/s & $\delta = 3 \times 10^{-3}$ m

$\tau = (1.97 \times 10^{-5})(10.8)\left[\frac{\pi}{2(3 \times 10^{-3})}\right]\cos\left(\frac{\pi(0)}{2(3 \times 10^{-3})}\right)$

$\tau_{wall} = 0.11$ Pa

For part (b), we basically want to find the altitude (y) where $\tau = \frac{1}{2}\tau_{wall}$, we found $\tau_{wall} = 0.11\ Pa$ in part a, so we just need to find y where $\tau = \frac{0.11}{2}\ Pa$. And our equation (1) works for the whole domain, we're just suppling the left hand side (τ) to find y in part b.

(b) $\tau = 0.11/2$

Sub. in (1),

$0.055 = \mu U\left(\frac{\pi}{2\delta}\right)\cos\left(\frac{\pi y}{2\delta}\right)$

$\therefore y = 2 \times 10^{-3}$ m

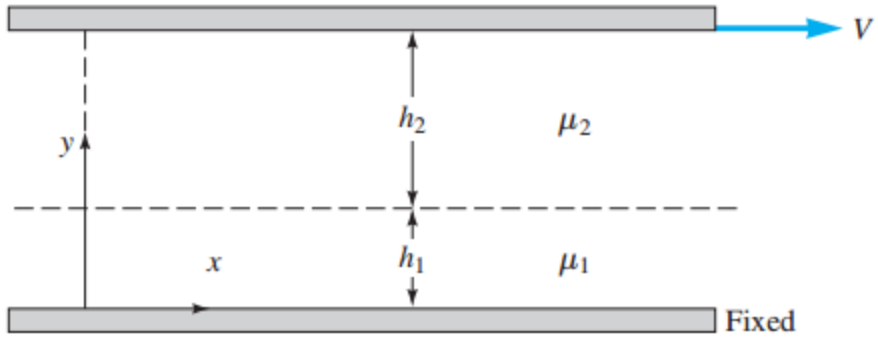
or $y = 2\frac{\delta}{3}$

The answer tells us that the position where τ is halved, is located about 2/3 way up the boundary layer (δ)

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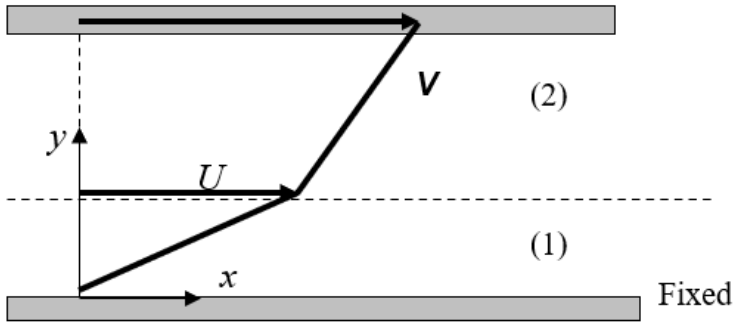
Problem 1

Examining the steady flow between a fixed lower plate and a moving upper plate, there are two immiscible liquids between the plates, as in Fig. P1.57.
(a) Sketch the expected no-slip velocity distribution u(y) between the plates.
(b) Find an analytic expression for the velocity U at the interface between the two liquid layers.
(c) What is the result of (b) if the viscosities and layer thicknesses are equal?



P1.57

Solution: We begin with the hint, from Fig. 1.8, that the shear stress is constant between the two plates. The velocity profile would be a straight line in each layer, with different slopes:



Ans. (a)

Here we have drawn the case where $\mu_2 > \mu_1$, hence the upper profile slope is less. (*b*) Set the two shear stresses equal, assuming no-slip at each wall:

$$\tau_1 = \mu_1 \frac{(U-0)}{h_1} = \tau_2 = \mu_2 \frac{(V-U)}{h_2}$$

$$\text{Solve for} \quad U = V \left[\frac{1}{1+\dfrac{\mu_1 h_2}{\mu_2 h_1}}\right] \qquad Ans.(b)$$

(*c*) For equal viscosities and layer thicknesses, we get the simple result $U = V/2$. *Ans.*

An immiscible liquid is incapable of mixing (e.g. oil always float above water)

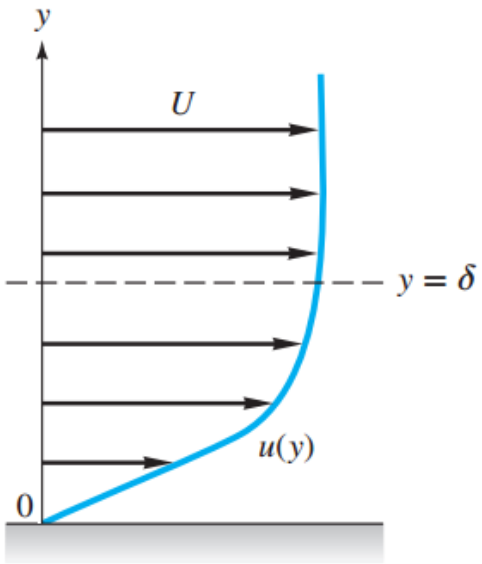
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Problem 2

P1.51 An approximation for the boundary-layer shape in Figs. 1.5*b* and P1.51 is the formula

$$u(y) \approx U \sin\left(\frac{\pi y}{2\delta}\right), \quad 0 \le y \le \delta$$

where U is the stream velocity far from the wall and δ is the boundary layer thickness, as in Fig. P1.51. If the fluid is helium at 20°C and 1 atm, and if $U = 10.8$ m/s and $\delta = 3\,mm$, use the formula to (*a*) estimate the wall shear stress τ_w in Pa, and (*b*) find the position in the boundary layer where τ is one-half of τ_w.



P1.51

P1.51 An approximation for the boundary-layer shape in Figs. 1.6*b* and P1.51 is the formula

$$u(y) \approx U \sin(\frac{\pi y}{2\delta}), \quad 0 \le y \le \delta$$

where U is the stream velocity far from the wall and δ is the boundary layer thickness, as in Fig. P.151. If the fluid is helium at 20°C and 1 atm, and if U = 10.8 m/s and δ = 3 mm, use the formula to (*a*) estimate the wall shear stress τ_w in Pa; and (*b*) find the position in the boundary layer where τ is one-half of τ_w.

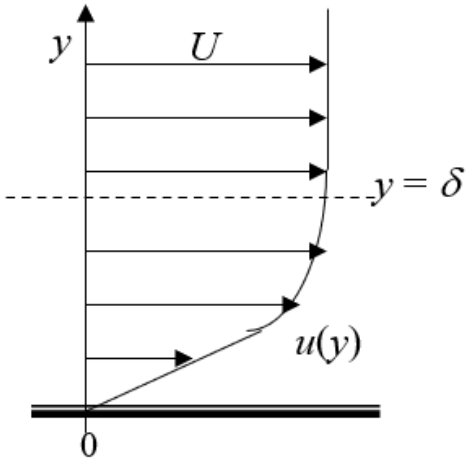


Fig. P1.51

Solution: From Table A.4, for helium, take $R = 2077$ m^2/(s^2-K) and $\mu = 1.97$E-5 kg/m-s.

(*a*) Then the wall shear stress is calculated as

$$\tau_w = \mu \frac{\partial u}{\partial y}\Big|_{y=0} = \mu(U \frac{\pi}{2\delta} \cos\frac{\pi y}{2\delta})_{y=0} = \frac{\pi \mu U}{2\delta}$$

$$\text{Numerical values: } \tau_w = \frac{\pi(1.97E-5\,kg/m-s)(10.8\,m/s)}{2(0.003\,m)} = \mathbf{0.11\ Pa} \quad Ans.(a)$$

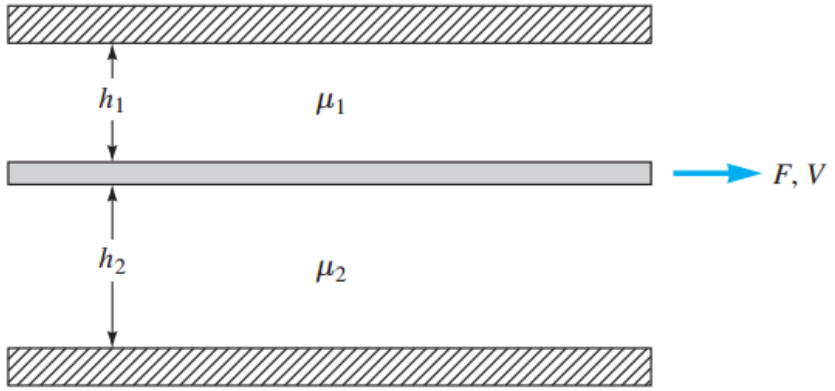
A very small shear stress, but it has a profound effect on the flow pattern.

(*b*) The variation of shear stress across the boundary layer is a cosine wave, $\tau = \mu\,(du/dy)$:

$$\tau(y) = \frac{\pi\mu U}{2\delta}\cos(\frac{\pi y}{2\delta}) = \tau_w \cos(\frac{\pi y}{2\delta}) = \frac{\tau_w}{2} \text{ when } \frac{\pi y}{2\delta} = \frac{\pi}{3}, \text{ or: } \mathbf{y = \frac{2\delta}{3}} \quad Ans.(b)$$

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Problem 3*



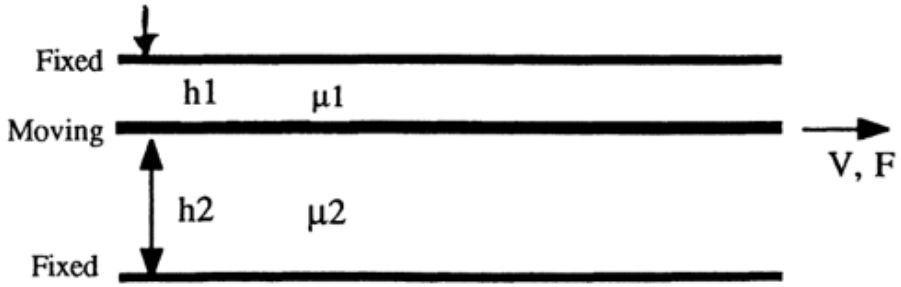
P1.48

A thin plate is separated from two fixed plates by very viscous liquids μ1 and μ2, respectively, as in Fig. P1.48. The plate spacings h1 and h2 are unequal, as shown. The contact area is A between the centre plate and each fluid.

(a) Assuming a linear velocity distribution in each fluid, derive the force F required to pull the plate at velocity V

(b) Is there a necessary relation between the two viscosities, μ1 and μ2 ?

P1.48 A thin moving plate is separated from two fixed plates by two fluids of unequal viscosity and unequal spacing, as shown below. The contact area is A. Determine (a) the force required, and (b) is there a necessary relation between the two viscosity values?



Solution: (a) Assuming a linear velocity distribution on each side of the plate, we obtain

$$F = \tau_1 A + \tau_2 A = \left(\frac{\mu_1 V}{h_1} + \frac{\mu_2 V}{h_2} \right) A \quad \textit{Ans.} \text{ (a)}$$

The formula is of course valid only for laminar (nonturbulent) steady viscous flow.

(b) Since the center plate separates the two fluids, they may have separate, unrelated shear stresses, and there is *no necessary relation* between the two viscosities.

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